The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 January 1994

INCH-POUND

MIL-S-19500/382B
4 October 1993
SUPERSEDING
MIL-S-19500/382A
1 July 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPE 2N2944A, 2N2945A, AND 2N2946A
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for low-power, high-speed chopper, PNP, silicon transistors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (similar to TO-46).

1.3 Maximum ratings.

Types	P_T 1/ T_A = +25°C	V_{EBO}	V_{CBO}	V_{CEO}	V_{ECO}	I_C	T_J and T_{STG}
	mW	V dc	V dc	V dc	V dc	mA dc	°C
2N2944A	400	-15	-15	-10	-10	-100	-65 to +200
2N2945A	400	-25	-25	-20	-20	-100	-65 to +200
2N2946A	400	-40	-40	-35	-35	-100	-65 to +200

1/ Derate linearly 2.3 mW/°C above T_A = +25°C.

1.4 Primary electrical characteristics.

Limits	h_{FE1} V_{CE} = -0.5 V dc I_C = -1 mA dc			h_{FE} (inv)1 V_{EC} = -0.5 V dc I_B = -200 µA dc			r_{ec} (on)2 I_e = 100 µA ac(rms) I_B = -1 mA dc I_E = 0; f = 1 kHz			C_{ibo} V_{EB} = -6 V dc I_C = 0 100 kHz ≤ f ≤ 1 MHz
	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A	
							ohms	ohms	ohms	pF
Min	100	70	50	50	30	20	---	---	---	---
Max	---	---	---	---	---	---	4	6	8	6

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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Limits	V_{EC}(ofs)						|hfe|			C_{obo}	
	I_B = -200 µA dc I_E = 0			I_R = -1.0 mA dc I_E = 0			f = 1 MHz V_{CE} = -6 V dc I_C = -1 mA dc			V_{CB} = -6 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	
	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A	2N2944A	2N2945A	2N2946A		
	mV dc	mV dc	mV dc	mV dc	mV dc	mV dc					pF
Min	---	---	---	---	---	---	15	10	5		---
Max	-0.3	-0.5	-0.8	-0.6	-1.0	-2.0	55	55	55		10

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

$V_{(BR)ECO}$ - - - Breakdown voltage, emitter to collector, with base open-circuited.

h_{FE} (inv) - - Forward-current transfer ratio except that the collector and emitter shall be interchanged, i.e., I_E/I_B.

I_e - - - - - - Emitter current (rms).

r_{ec} (on)- - - Small-signal emitter-collector on-state resistance.

V_{EC} (ofs) - - Emitter to collector offset voltage, i.e., open-circuit voltage between emitter collector when the base-collector junction is forward-biased.

V_{ec} - - - - - Emitter to collector voltage (rms).

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V_{EC} (ofs) - - Emitter to collector offset voltage, i.e., open-circuit voltage between emitter collector when the base-collector junction is forward-biased.

V_{ec} - - - - - Emitter to collector voltage (rms).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 9 (T-28).

3.3.1 Lead finish. Where a choice of lead finish is desired, it shall be specified in the acquisition requirements (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS	JAN, JANTX, and JANTXV
9	I_{CBO1} and h_{FE} $(inv)_1$	Not applicable
11	I_{CBO1}; h_{FE} $(inv)_1$; ΔI_{CBO1} = 100 percent of initial value; Δh_{FE} $(inv)_1$ = 25 percent of initial value.	I_{CBO1} and h_{FE} $(inv)_1$
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100 percent of initial value or .2 nA dc for 2N2944A and 2N2945A, .5 nA dc for 2N2946A; Δh_{FE} $(inv)_1$ = 25 percent of initial value.	Subgroup 2 of table I herein, ΔI_{CBO1} = 100 percent of initial value or .2 nA dc for 2N2944A and 2N2945A, .5 nA dc; Δh_{FE} $(inv)_1$ = 25 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = Room ambient as defined in 4.5 of MIL-STD-750
P_T = +400 mW
2N2944A: V_{CB} = -8 V dc
2N2945A: V_{CB} = -16 V dc
2N2946A: V_{CB} = -28 V dc

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B3	2037	Condition A
B4	1037	P_T = 400 mW; T_A = +25°C ±3°C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling on the devices shall be permitted. V_{CB} = -8 V dc (2N2944A); V_{CB} = -16 V dc (2N2945A); V_{CB} = -28 V dc (2N2946A).
B5	1027	T_A = +125°C ±25°C; for 96 hours; P_T = 400 mW at +100°C or adjusted as required by the chosen T_A to give an average lot. T_J = +275°C. Marking legibility requirements shall not apply; V_{CB} = -8 V dc (2N2944A); V_{CB} = -16 V dc (2N2945A); V_{CB} = -28 V dc (2N2946A).

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	P_T = 400 mW; T_A = +25°C ±3°C; V_{CB} = -8 V dc (2N2944A); V_{CB} = -16 V dc (2N2945A); V_{CB} = -28 V dc (2N2946A).
B3	2037	Test condition A.

4.4.4 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	P_T = 400 mW; T_A = +25°C ±3°C; V_{CB} = -8 V dc (2N2944A); V_{CB} = -16 V dc (2N2945A); V_{CB} = -28 V dc (2N2946A).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Input capacitance. This test shall be conducted in accordance with MIL-STD-750, method 3240, except the output capacitor shall be omitted.

4.5.2 Emitter to collector breakdown voltage. Method of test shall be in accordance with method 3011 of MIL-STD-750, condition D, except that all references to the collector and the emitter of the transistor shall be interchanged.

4.5.3 Forward-current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and the emitter of the transistor shall be interchanged. Then:

$$h_{FE}\ (inv) = \frac{I_E}{I_B}$$

4.5.4 Emitter to collector offset voltage. The transistor shall be tested in the circuit shown on figure 1. The base current shall be adjusted to the specified value. The voltage between the emitter and collector shall then be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

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4.5.5 Small-signal emitter-collector on-state resistance. The transistor shall be tested in the circuit shwon on figure 2. The base current shall be adjusted to the specified value and an ac sinusoidal signal current, I_e, of the specified rms value shall be applied between the emitter and collector. The rms voltage, V_{ec}, between the emitter and collector shall be measured using an ac voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The small-signal emitter-collector on-state resistance shall then be determined as follows:

$$r_{EC}\ (on) = \frac{V_{EC}}{I_E}$$

Where: V_{ec} is the rms voltage between the emitter and collector.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = -10 µA dc	$V_{(BR)CEO}$			
2N2944A				-10		V dc
2N2945A				-20		V dc
2N2946A				-35		V dc
Breakdown voltage collector to base	3001	Bias condition D; I_C = -10 µA dc	$V_{(BR)CBO}$			
2N2944A				-15		V dc
2N2945A				-25		V dc
2N2946A				-40		V dc
Breakdown voltage emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$			
2N2944A				-15		V dc
2N2945A				-25		V dc
2N2946A				-40		V dc
Breakdown voltage, emitter to collector	3011	Bias condition B; I_E = -10 µA dc; I_B = 0; (see 4.5.2)	$V_{(BR)ECO}$			
2N2944A				-10		V dc
2N2945A				-20		V dc
2N2946A				-35		V dc
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
2N2944A		V_{CB} = -15 V dc			-0.1	nA dc
2N2945A		V_{CB} = -25 V dc			-0.2	nA dc
2N2946A		V_{CB} = -40 V dc			-0.5	nA dc
Emitter to base cutoff current	3061	Bias condition D	I_{EBO1}			
2N2944A		V_{EB} = -15 V dc			-0.1	nA dc
2N2945A		V_{EB} = -25 V dc			-0.2	nA dc
2N2946A		V_{EB} = -40 V dc			-0.5	nA dc
Forward-current transfer ratio	3076	V_{CE} = -0.5 V dc; I_C = -1.0 mA dc	h_{FE1}			
2N2944A				100		
2N2945A				70		
2N2946A				50		

See footnote at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_B = -200 µA dc; (see 4.5.3)	h_{FE} (inv)$_1$			
2N2944A				50		
2N2945A				30		
2N2946A				20		
Emitter to collector offset voltage		I_B = -200 µA dc; I_E = 0; (see 4.5.4 and figure 1).	V_{EC} (ofs)$_1$			
2N2944A					-0.3	mV dc
2N2945A					-0.5	mV dc
2N2946A					-0.8	mV dc
Emitter to collector offset voltage		I_B = -1 mA dc; I_E = 0; (see 4.5.4 and figure 1).	V_{EC} (ofs)$_2$			
2N2944A					-0.6	mV dc
2N2945A					-1.0	mV dc
2N2946A					-2.0	mV dc
Emitter to collector offset voltage		I_B = -2 mA dc; I_E = 0; (see 4.5.4 and figure 1).	V_{EC} (ofs)$_3$			
2N2944A					-1.0	mV dc
2N2945A					-1.6	mV dc
2N2946A					-2.5	mV dc
Subgroup 3						
High-temperature operation:		T_A = +100°C				
Collector to base cutoff current	3036	Bias condition D; I_C = 0	I_{CBO2}			
2N2944A		V_{CB} = -15 V dc			-10	nA dc
2N2945A		V_{CB} = -25 V dc			-20	nA dc
2N2946A		V_{CB} = -40 V dc			-25	nA dc
Emitter to base cutoff current	3061	Bias condition D; I_C = 0	I_{EBO2}			
2N2944A		V_{CB} = -15 V dc			-10	nA dc
2N2945A		V_{CB} = -25 V dc			-15	nA dc
2N2946A		V_{CB} = -40 V dc			-20	nA dc

See footnote at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued.						
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = -0.5 V dc; I_C = -1 mA dc	h_{FE2}			
2N2944A				35		
2N2945A				25		
2N2946A				20		
Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_B = -200 µA dc; (see 4.5.3)	h_{FE} (inv)$_2$			
2N2944A				25		
2N2945A				15		
2N2946A				10		
Subgroup 4						
Small-signal emitter-collector on-state resistance		I_B = -100 µA dc; I_E = 0; I_e = 100 µA ac (rms); f = 1 kHz; (see 4.5.5 and figure 2)	r_{ec} (on)$_1$			
2N2944A					10	Ω
2N2945A					12	Ω
2N2946A					14	Ω
Small-signal emitter-collector on-state resistance		I_B = -1 mA dc; I_E = 0; I_e = 100 µA ac (rms); f = 1 kHz; (see 4.5.5 and figure 2)	r_{ec} (on)$_2$			
2N2944A					4	Ω
2N2945A					6	Ω
2N2946A					8	Ω
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio		V_{CE} = -6 V dc; I_C = -1 mA dc; f = 1 MHz	h_{fe}			
2N2944A				15	55	
2N2945A				10	55	
2N2946A				5	55	
Open circuit output capacitance	3236	V_{CB} = -6 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		10	pF
Input capacitance (output open-circuited)	3240	V_{EB} = -6 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz; (see 4.5.1)	C_{ibo}		6.0	pF

See footnote at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Pulse response:						
Delay time	3251	Test condition B; (see figure 3)	t_d		50	ns
Rise time	3251	Test condition B; (see figure 3)	t_r		100	ns
Storage time	3251	Test condition B; (see figure 3)	t_s		350	ns
Fall time	3251	Test condition B; (see figure 3)	t_f		100	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Collector to base cutoff current	3036	Bias condition D; I_E = 0	I_{CBO1}			
	2N2944A		V_{CB} = -15 V dc			-0.2	nA dc
	2N2945A		V_{CB} = -25 V dc			-0.4	nA dc
	2N2946A		V_{CB} = -40 V dc			-1.0	nA dc
2.	Collector to base cutoff current	3036	Bias condition D; I_E = 0	I_{CBO1}			
	2N2944A		V_{CB} = -15 V dc			-0.5	nA dc
	2N2945A		V_{CB} = -25 V dc			-1.0	nA dc
	2N2946A		V_{CB} = -40 V dc			-2.5	nA dc
3.	Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_B = -200 µA dc; (see 4.5.3)	h_{FE} (inv)$_1$			
	2N2944A				50		
	2N2945A				30		
	2N2946A				20		
4.	Emitter to collector offset voltage		I_B = -200 µA dc; I_E = 0; (see 4.5.4 and figure 1)	V_{EC} (ofs)$_1$			
	2N2944A					-0.3	mV dc
	2N2945A					-0.5	mV dc
	2N2946A					-0.8	mV dc
5.	Collector to base cutoff current	3036	Bias condition D; I_E = 0	ΔI_{CBO1}	100 percent of initial value		
	2N2944A		V_{CB} = -15 V dc				
	2N2945A		V_{CB} = -25 V dc				
	2N2946A		V_{CB} = -40 V dc				
6.	Forward-current transfer ratio (inverted connection)	3076	V_{EC} = -0.5 V dc; I_B = 200 µA dc; (see 4.5.3)	Δh_{FE} (inv)$_1$	25 percent of initial value		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroups 3 and 4, see table II herein, steps 1, 3, and 4.

b. Subgroup 5, see table II herein, steps 1, 3, 4, 5, and 6.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1, 3, and 4.

b. Subgroups 3 and 6, see table II herein, steps 2, 3, 4, and 6.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, steps 1, 3, and 4.

b. Subgroup 6, steps 1, 3, 4, 5, and 6 (JANS) and 2, 3, 4, and 6 (JAN, JANTX, and JANTXV).

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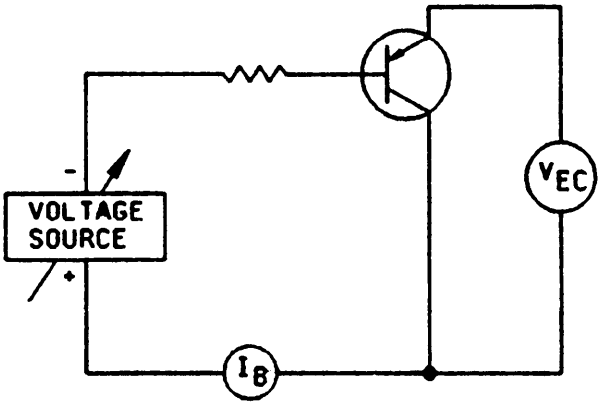


FIGURE 1. Emitter to collector offset voltage test circuit.

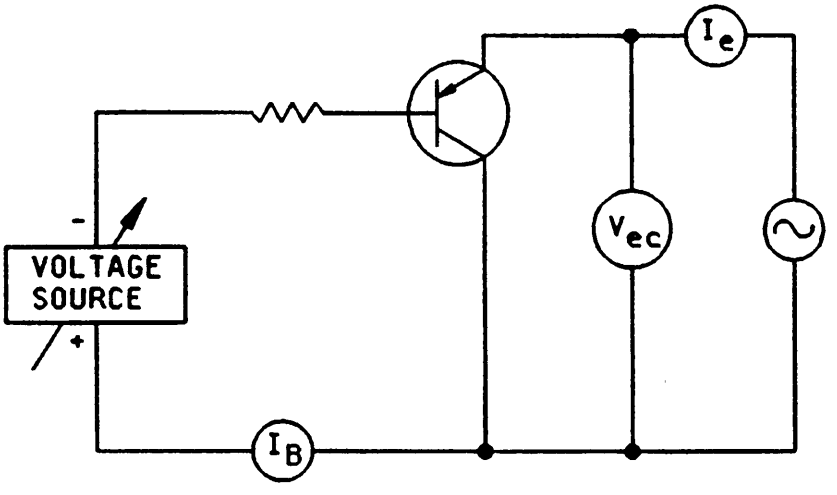
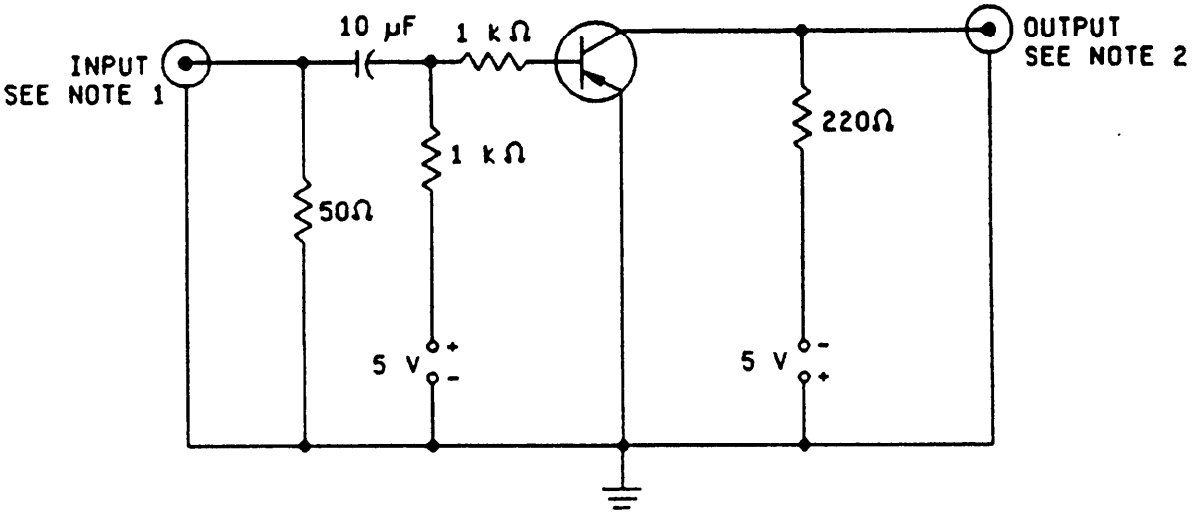
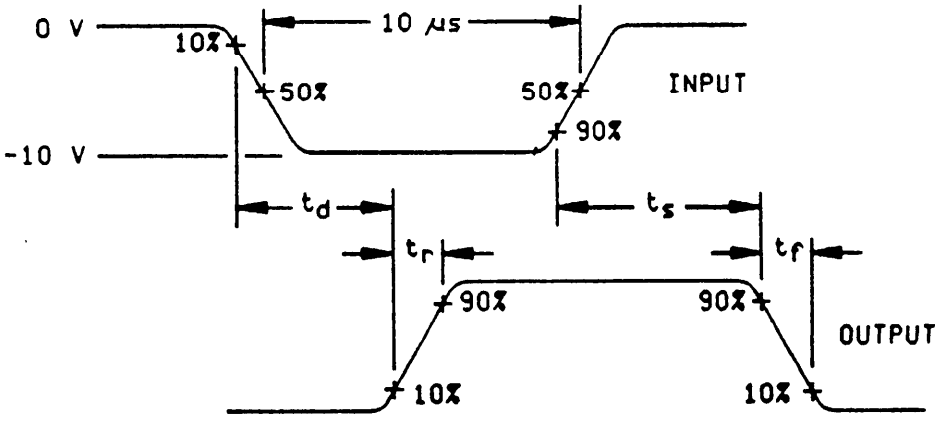


FIGURE 2. Emitter to collector offset voltage test circuit.

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TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be ≤ 10 ns, duty cycle ≤ 2 percent. The input pulse width shall be 500 ns.
2. Output monitored with an oscilloscope with the following characteristics: Z_{in} ≥ 1 MΩ, t_r ≤ 1 ns.

FIGURE 3. Pulse response test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish as specified (see 3.3.1).

d. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AV
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/382B	2. DOCUMENT DATE (YYMMDD) 93/10/04

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER TYPE 2N2944A, 2N2945A, AND 2N2946A JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) NASA/PARTS PROJECT OFFICE (NPPO) NASA GODDARD SPACE FLIGHT CENTER CODE 310.A GREENBELT, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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